

Ex 6: Programs in Prolog

1. Family Relationships

Problem: Define family relations and answer grandparent queries.

% Facts

parent(john, mary).

parent(mary, susan).

parent(john, mike).

% Rules

grandparent(X, Y) :- parent(X, Z), parent(Z, Y).

OUTPUT:

```
26 ?- consult(['c:/Users/Lenovo/Downloads/poai6.1.pl']).
true.

27 ?- grandparent(john, susan).
true
```

2: Animal Classification

Problem: Identify if an animal is a mammal based on features.

% Facts

has_hair(tiger).

gives_milk(tiger).

% Rule

mammal(X) :- has_hair(X), gives_milk(X).

OUTPUT:

```
29 ?- consult(['c:/Users/Lenovo/Downloads/poai6.2.pl']).
true.

30 ?- mammal(tiger).
true.

31 ?- ◆
```

3: Simple Arithmetic Knowledge Base

Problem: Implement simple addition using PROLOG.

% Rule

add(X, Y, Z) :- Z is X + Y.

OUTPUT:

```
31 ?- consult(['c:/Users/Lenovo/Downloads/poai6.3.pl']).  
true.
```

```
32 ?- add(5, 7, Z).  
Z = 12.
```

```
33 ?- ◆
```